Artemis Momentum Factor

Research Report — Stage 5 Deliverable

Date: 2026-05-30 | Spot long/short, Artemis-sourced universe | Survivorship-corrected, cost-aware, sealed out-of-sample

Provenance: the full Stage-2 table is read from `data/stats/significance.parquet`; narrative, Stage-4, audit, and disclosure values are verified literals from the committed source docs. This build does not recompute statistics.

VERDICT: NO-DEPLOY

1. Conclusion first

We tested cross-sectional momentum on an Artemis-sourced crypto spot universe and find **no deployable factor**. The recommendation is: **do not deploy**.

- **What we tested.** Twenty-one momentum variants — a pre-registered selection family of **7 lookbacks** (1, 3, 5, 7, 14, 28, 56 days) at the convention skip of 1 day, plus 14 skip-{2,3} **diagnostics** reported for transparency but never eligible for deployment. The full Stage-2 battery (Newey-West HAC, Lo Sharpe SE, spanning vs market + size control, stationary block bootstrap, Bonferroni / HLZ, subsample sign-stability, DSR and PBO) ran on the in-sample slice; a sealed OOS window was opened exactly once.
- **What held up: nothing, at the family level.** No selection-family variant is significant on Newey-West against the Bonferroni threshold (0.05/7 = 0.00714). The strongest, `momentum_L5d_S1d`, is HLZ-"suggestive" with HAC t = 2.40 and HAC p = 0.0081 — *above* the threshold. It clears Bonferroni **only** via the bootstrap-override rule (bootstrap p = 0.0058), and is then **disqualified** by the Stage-2.6 deployment gate for flipping sign across subsamples. DSR = 0.46; grid PBO = 0.114.
- **Deployed-candidate net-of-cost OOS Sharpe.** Characterized for completeness, `momentum_L5d_S1d` posts OOS **net Sharpe 0.756** (gross 0.897) and IS net Sharpe 0.664 (gross 0.789). The OOS Sharpe did **not** collapse — but this is a **single-regime artifact** on ~30 overlapping observations spent exactly once (the 2024 crypto bull), not a stable edge. The academic 4-week comparator `momentum_L28d_S1d` **works only gross of costs**: gross Sharpe 0.073 turns to **net -0.031** in-sample and **net -0.270** out-of-sample.
- **Recommendation.** Do not deploy momentum on the Artemis spot universe. The suggestive ~5-day signal is a *research lead*, not a strategy. Capacity (~$36M) is comfortable and is **not** the binding constraint; the disqualification rests on multiple-testing failure, subsample sign-instability, regime dependence, lookback fragility, and disclosed data limits.

2. Methodology (brief)

Data — Artemis only, survivorship-corrected. The universe is enumerated programmatically from the Artemis `/asset` catalog (1,013 entries; **846 with any price data**), on a daily point-in-time grid spanning **2018-01-01 to 2026-05-30** (2,598,912 panel rows). Eligibility filters (≥90 days history, ≥50% observation density, $10M market-cap floor, trailing-30d median 24H volume above threshold, staleness check, stablecoin and wrapped-token exclusion) are rebuilt as-of each date. Dead and collapsed coins are **kept**: **238 of 846 (28%)** exhibit a terminal >90% drawdown (15 delisted + 223 still-printing zombies), and every crash return is carried into the P&L (the LUNA/`lunc` -99.99% collapse included). Recycled tickers are split into distinct synthetic assets. Residual survivorship — coins purged from Artemis before the query — is **unestimable and disclosed**; reported factor returns are **upper bounds**.

Returns. Holding return is the simple spot price return — Artemis exposes no perpetual funding, so **no funding term** is modeled (disclosed). Simple returns aggregate across coins; log returns compound through time; the two are never mixed.

Momentum construction. Signal = trailing log-sum return over a fixed lookback, shifted by the skip. The **7-lookback grid is fixed in advance** (academic 14/28/56-day plus crypto-short 1/3/5/7-day); skip is fixed by convention at **1 day**. The eligible universe is sorted into **quintiles** (long top 20% / short bottom 20%, equal-weight, dollar-neutral). Signal uses data through close of t; position is entered at the **close of t+1** — a mandatory, unit-tested one-period lag (no look-ahead).

Stage-2 significance battery. Per variant: naive t-stat (reported but flagged biased, never the headline); **Newey-West HAC t-stat** (the reported mean-return test, bandwidth covering holding-period overlap); Sharpe with the **Lo (2002) SE** (autocorrelation-corrected when Ljung-Box fires); a **spanning regression** on {equal-weight market, small-minus-big size control} reporting the HAC-t alpha (size control is TEST-ONLY, never deployed); **Bonferroni** on the family of 7 and HLZ tiers; a **stationary block bootstrap** (arch) as the bootstrap of record, which overrides Newey-West on a cross-threshold disagreement; **subsample sign-stability** (a sign-flip disqualifies regardless of t-stat); and **DSR** plus **PBO/CSCV** for selection overfit.

Sealed OOS and cost-aware backtest. The most recent ~30% of rebalance dates (`OOS_START = 2023-12-02`) were reserved before any statistic was computed and opened **exactly once** under a single-use guard. The backtest applies a 10 bps spot taker fee per side plus size-scaled tiered slippage, executes at the t+1 close, and reports the full net metric set plus a capacity estimate.

3. Factor-by-factor (variant) results

None of the selection family (skip = 1) is significant on the Newey-West HAC test at the Bonferroni threshold (0.05/7 = 0.00714). The diagnostics (skip = 2,3) are reported separately and were **never eligible for deployment**; the striking diagnostic Sharpes (L3d/S3d HAC t ≈ 5.0, L14d/S3d HAC t ≈ 3.9) are **not** in the selection family. The full numeric table for all 21 variants is in the Appendix (7.3).

- `L5d/S1d` (strongest): HAC t = 2.403, HAC p = 0.0081 (> 0.00714), bootstrap p = 0.0058, suggestive, span α t = 2.211, DSR = 0.460. **Survives Bonferroni only via the bootstrap override, and holds_sign = NO — DISQUALIFIED.**
- `L7d/S1d`: HAC t = 1.887, p = 0.0296 — not significant. `L14d/S1d`: HAC t = 1.945, p = 0.0259 — not significant. `L3d/S1d`: HAC t = 1.533, sign-unstable.
- `L28d/S1d` (academic 4-week): HAC t = 0.955, p = 0.1698 — not significant. `L56d/S1d`: HAC t = 0.756 — not significant. `L1d/S1d`: HAC t = -1.300 (negative; reversal), sign-unstable.
- Grid overfit controls: **PBO/CSCV = 0.114** (below the 0.5 threshold) but on a correlated family (Fig 5), so the effective independent-test count is below 7.

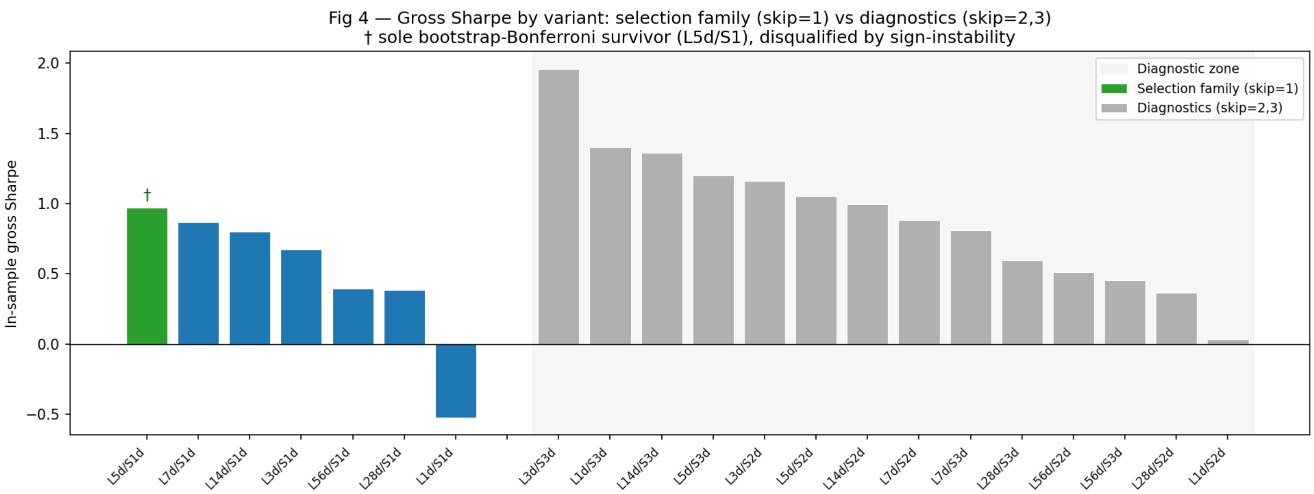

Fig 4 — Gross Sharpe by variant (selection family vs diagnostics). Takeaway: No selection-family variant clears the Bonferroni threshold on Newey-West; the eye-catching high Sharpes (L3d/S3, L14d/S3) are in the diagnostic zone and were never eligible for deployment.

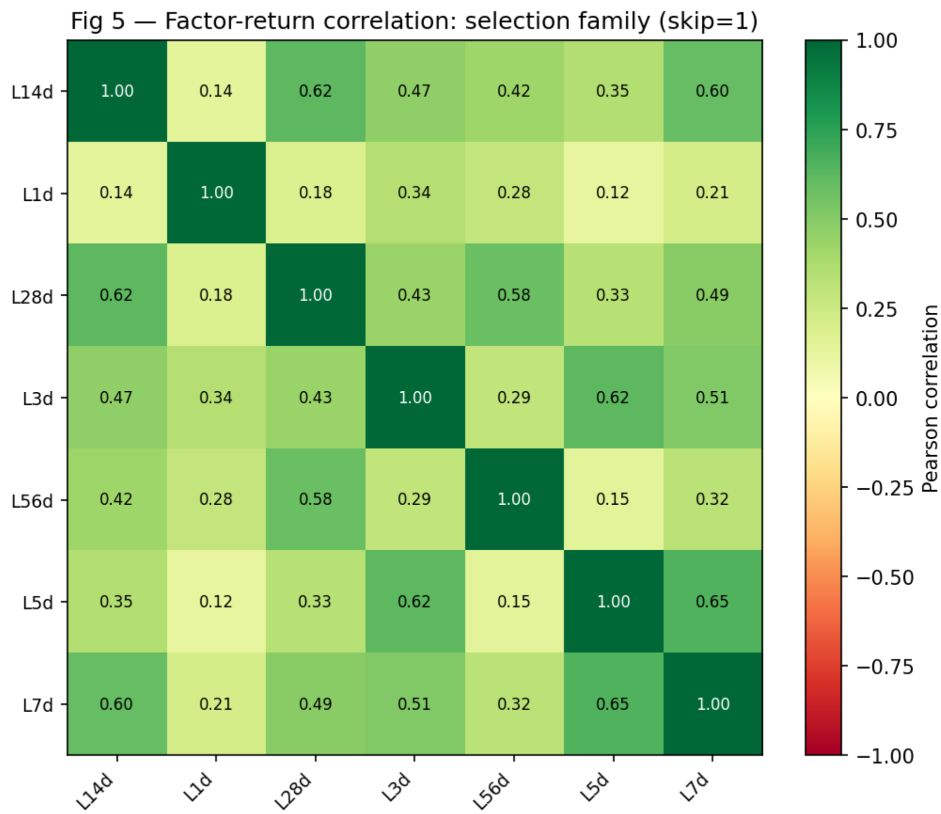

Fig 5 — Factor-return correlation (selection family). Takeaway: Short lookbacks (L1d-L5d) cluster together with high mutual correlation, while longer lookbacks (L14d-L56d) form a separate cluster; the family is far from orthogonal, reducing the independent-test count below 7.

4. Deployed-candidate characterization (Stage-4 backtest)

For completeness we characterized the strongest candidate, `momentum_L5d_S1d`, net of realistic spot costs. **This characterizes the candidate; it does not rescue it** — the Stage-2 disqualification stands.

Gross vs net, in-sample vs out-of-sample (primary L5d/S1d):

segment	gross Sharpe	net Sharpe	net ann ret	net ann vol	Sortino	max DD	Calmar	hit rate	ann turnover
in-sample	0.789	0.664	0.2490	0.3752	0.756	-0.4851	0.513	0.574	24.31
out-of-sample	0.897	0.756	0.2469	0.3266	1.031	-0.2516	0.981	0.567	28.22

Gross and net are shown side by side; net is never flattered above gross. IS and OOS are shown side by side; the OOS window was opened exactly once.

- **IS – OOS net Sharpe gap = -0.092.** The OOS figure exceeds in-sample, but this is a spent-once, ~30-observation, single-regime (2024-bull) artifact — not evidence of a stable edge.
- **Capacity ≈ $36,044,481** — the AUM at which size-scaled slippage on the actual per-rebalance traded order (~2.0× summed one-way turnover) erases the gross edge (per-rebalance gross edge 0.02436). Comfortably above a $1M book, so **capacity is not the binding constraint**.
- **Turnover** ~24 (IS) / ~28 (OOS) annualized — high, consistent with a 5-day signal.

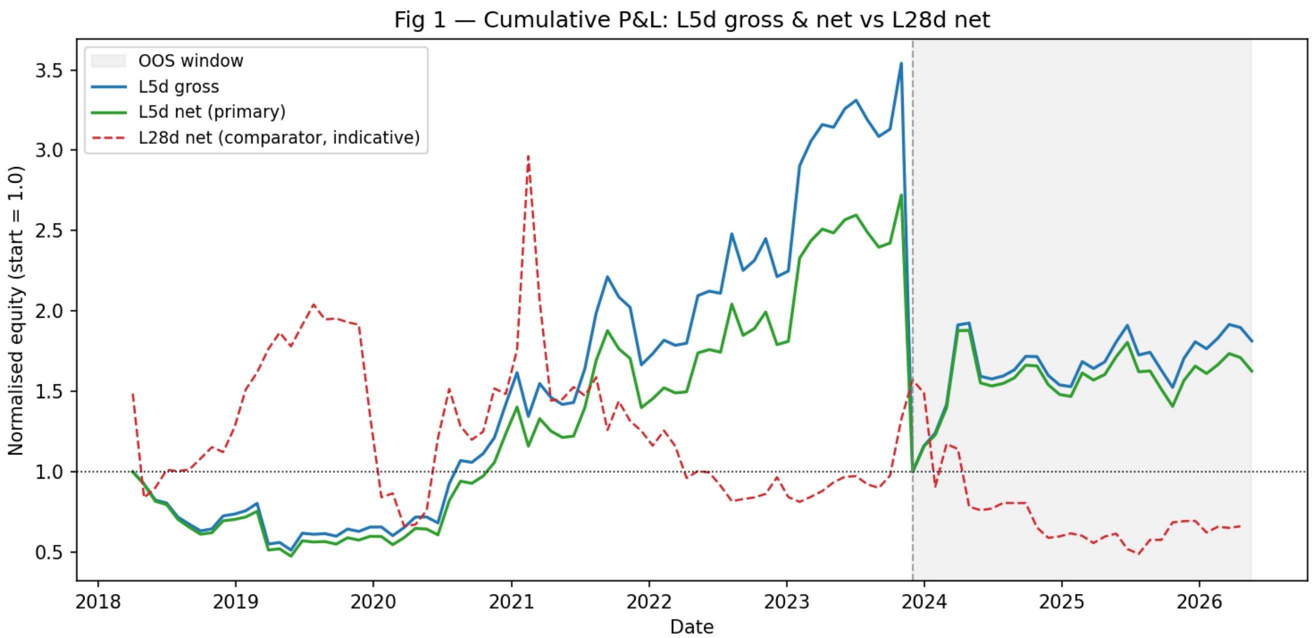

Fig 1 — Cumulative P&L; (L5d gross & net, L28d net). Takeaway: L5d net grows steadily in-sample but is a regime-sensitive path; L28d net decays to near-zero — the canonical 4-week lookback does not survive costs.

5. Robustness

2× costs (in-sample, primary): net Sharpe degrades from **0.664** → **0.552** (gross unchanged at 0.789; max DD widens to -0.5219). The edge thins materially under a doubled cost assumption.

±50% lookback (in-sample, net) — the construction is fragile to its one free parameter:

variant	net Sharpe	net ann ret	note
momentum_L5d_S1d (chosen)	0.664	0.2490	chosen lookback = 5d
momentum_L2d_S1d	-0.328	-0.1155	lookback-50% (skip/quantile unchanged)
momentum_L8d_S1d	0.621	0.2645	lookback+50% (skip/quantile unchanged)

A -50% lookback flips the candidate net-negative — not robust.

Regime breakdown (in-sample, net mean return per regime):

regime	n	mean net return
bull	17	-0.00624
bear	28	0.02195
chop	22	0.03453

Caveat (do not over-read): this is a full-sample, **descriptive** partition, **not** a walk-forward signal — it could not have been traded ex-ante. The `chop` bucket is the top-|market-move| tercile, which on this sample skews toward large up moves, so the apparent ‘negative in bull / positive in bear’ contrast is **overstated**. The disqualifying evidence is the Stage-2 §2.6 sign-instability itself, not this descriptive split.

One-shot OOS, single-regime character. The OOS window is 30 overlapping-regime observations spent exactly once; a single favorable stretch (the 2024 crypto bull) carries the positive OOS Sharpe. A near-zero or negative OOS Sharpe would have been reported as overfitting; the positive figure is reported as-is and interpreted as regime exposure, consistent with the sign-flip disqualification.

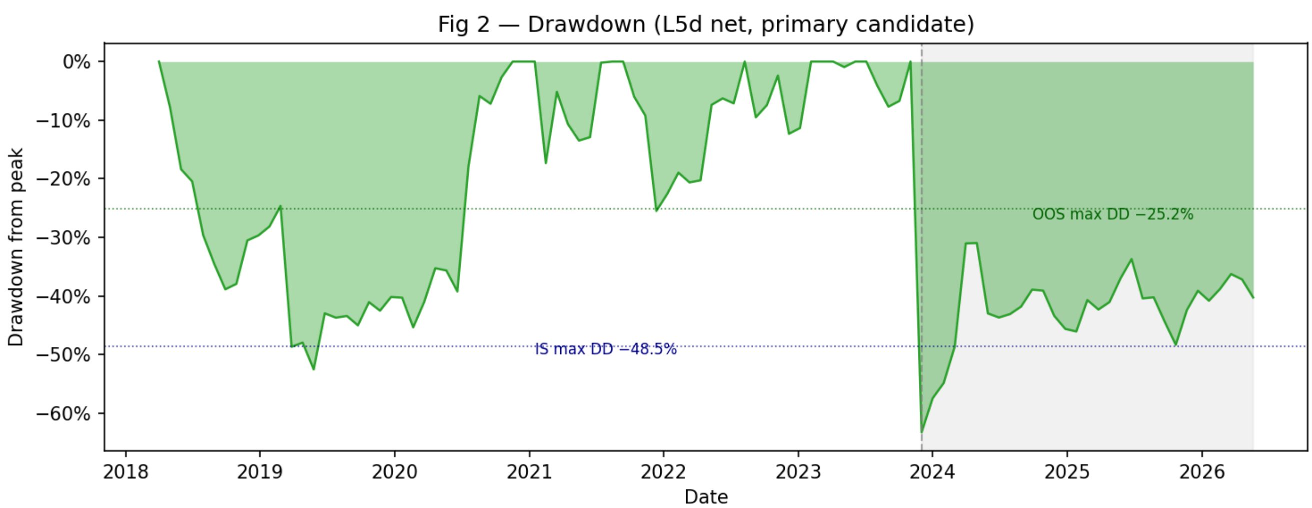

Fig 2 — Drawdown (L5d net, primary). Takeaway: In-sample max drawdown of -48.5% dwarfs the OOS -25.2%, consistent with the sign-instability finding; drawdown is material relative to returns at both horizons.

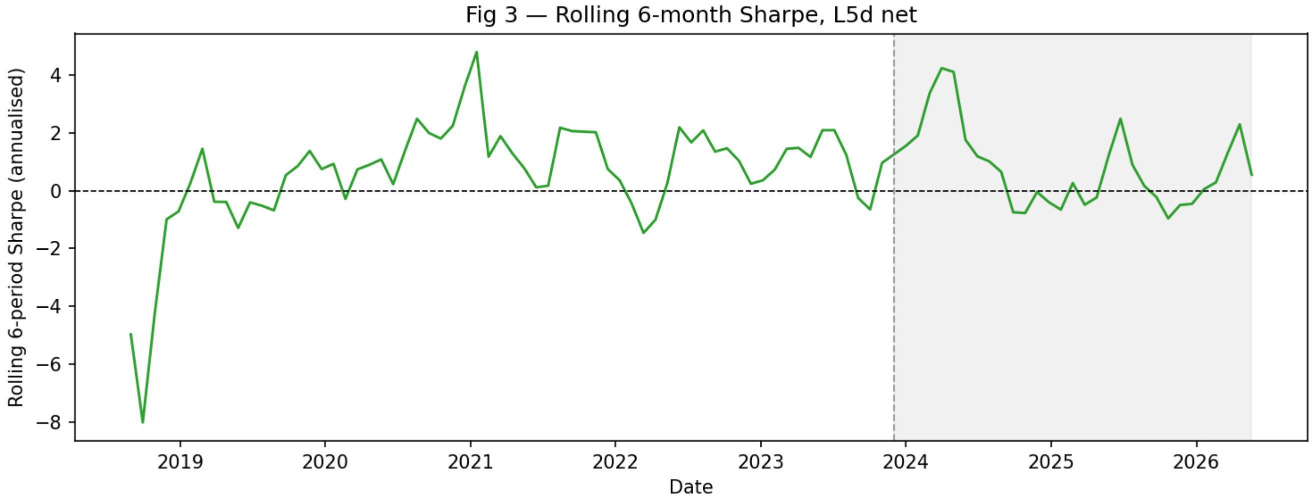

Fig 3 — Rolling 6-month Sharpe (L5d net). Takeaway: Rolling Sharpe oscillates widely and dips sharply negative in multiple sub-windows, confirming that any positive in-sample/OOS Sharpe is driven by episodic regimes rather than a stable factor.

6. Recommendation

Do not deploy. Target allocation to a momentum sleeve on the Artemis spot universe: **zero**.

- **Expected Sharpe (range):** for the strongest candidate, net Sharpe sits in roughly **0.66–0.76** in the realized sample, but with **no statistically reliable edge** and a true expected Sharpe indistinguishable from zero once multiple testing, sign-instability, and the single-regime OOS are accounted for. We do not represent this as a deployable expectation.
- **The suggestive ~5-day signal is a research lead, not a strategy.** It would warrant follow-up only with (a) more independent observations, (b) a cleaner, lower-turnover construction, and (c) a true point-in-time universe addressing residual survivorship.
- **Key risks if (against this recommendation) deployed:** regime dependence (return is regime exposure, not a stable factor); selection / multiple-testing fragility (no family Bonferroni survivor on its own terms); subsample **sign-instability** (the hard disqualifier); parameter fragility (±50% lookback breaks it); and data limitations (no funding, daily t+1-close execution, volume unreliability, residual survivorship).
- **Next steps:** treat momentum as closed for deployment on this universe; if revisited, pursue the three conditions above, and consider it only as one input to a broader, properly multiple-testing-controlled research program — never as a standalone sleeve.

7. Appendix — full tables and required disclosures

7.1 Required disclosures (spec §5.4)

• **Gross vs net side by side** (Section 4): spot costs reduce every reported Sharpe; net is never flattered above gross.

• **In-sample vs out-of-sample side by side** (Section 4): the OOS window was opened exactly once (single-use guard); a near-zero / negative OOS Sharpe would be reported as overfitting, not hidden.

• **'Works only gross' — momentum_L28d_S1d:** the academic 4-week canonical lookback is positive gross (IS gross Sharpe 0.073) but **net-negative** both in-sample (**net -0.031**) and out-of-sample (**net -0.270**, gross -0.158). The canonical horizon does not work net of costs.

• **No funding / spot:** Artemis exposes no perpetual funding; returns and costs carry no funding term (the guide's third cost component is N/A and stated).

• **Daily, t+1-close execution:** Artemis serves daily granularity only; fills are at the t+1 close with slippage applied — a one-period lag, not a costless intraday fill.

• **Wrapped-token exclusion:** wrapped tokens (WBTC, WETH, stETH, …) are excluded as redundant price exposures — a deliberate, disclosed deviation.

• **Residual (as-of-today catalog) survivorship:** the Artemis catalog is an as-of-today snapshot with no listing/delisting dates; coins fully purged before the query are unrecoverable — an **unestimable lower bound** on survivorship bias. Reported factor returns are **upper bounds**.

• **Volume reliability:** only 24H_VOLUME is historical (30D_VOLUME is real-time only) and is unreliable cross-sectionally; liquidity is gated on a market-cap floor + trailing-30d **median** 24H volume — a disclosed deviation from the mean-ADV rule.

• **Survivorship flows into P&L:** a collapsed short-leg coin's crash books as a positive contribution; dead coins are **not** dropped (238/846, 28% terminal collapses carried).

7.2 Universe / survivorship figures (docs/AUDIT.md)

Metric	Value
Artemis /asset catalog entries (as-of-today)	1,013
Assets with any price data (panel)	846
Date range	2018-01-01 → 2026-05-30
Panel rows (date × symbol)	2,598,912
Total terminal collapses (>90% drawdown, carried)	238 (28%)
— delisted cohort	15
— zombie cohort (still printing near-zero)	223
Left-censored assets (first price = 2018-01-01)	272
Assets ever eligible	303
Assets eligible on latest date	226

7.3 Full Stage-2 significance table — all 21 variants, INCLUDING failures

Rows are the pre-registered selection family (fam = sel, skip = 1) first, then the skip-{2,3} diagnostics (fam = diag). The HAC (Newey-West) t-stat is the reported mean-return test; the naive t is shown but biased. holds_sign = NO is a hard deployment disqualifier. The highlighted row is the sole bootstrap-Bonferroni survivor (L5d/S1d), itself disqualified by sign-instability. Numbers are verbatim from data/stats/significance.parquet.

variant	fam	n	naive t	HAC t	HAC p	boot p	HLZ	ann ret	Sharpe (SE)	span α	span α t	holds sign	DSR	survives
L14d/S1d	sel	69	1.902	1.945	0.0259	0.0354	n.s.	0.5128	0.799 (0.425)	0.02343	1.605	yes	0.308	no
L1d/S1d	sel	69	-1.247	-1.300	0.9032	0.9102	n.s.	-0.3078	-0.524 (0.422)	-0.02494	-1.450	NO	0.000	no
L28d/S1d	sel	69	0.911	0.955	0.1698	0.1780	n.s.	0.2791	0.383 (0.421)	-0.00166	-0.088	yes	0.060	no
L3d/S1d	sel	69	1.606	1.533	0.0626	0.0656	n.s.	0.4098	0.674 (0.424)	0.01286	0.822	NO	0.198	no

variant	fam	n	naive t	HAC t	HAC p	boot p	HLZ	ann ret	Sharpe (SE)	span α	span α t	holds sign	DSR	survives
L56d/S1d	sel	69	0.935	0.756	0.2249	0.2024	n.s.	0.2683	0.393 (0.525)	0.00798	0.360	NO	0.062	no
L5d/S1d	sel	69	2.315	2.403	0.0081	0.0058	suggestive	0.6163	0.972 (0.428)	0.04651	2.211	NO	0.460	YES
L7d/S1d	sel	69	2.068	1.887	0.0296	0.0214	n.s.	0.6209	0.868 (0.426)	0.04047	1.621	yes	0.352	no
L14d/S2d	diag	69	2.369	2.298	0.0108	0.0140	suggestive	0.6492	0.995 (0.428)	0.03692	2.205	yes	0.481	no
L14d/S3d	diag	69	3.241	3.949	0.0000	0.0006	sig	0.8988	1.361 (0.436)	0.06076	3.892	yes	0.805	no
L1d/S2d	diag	69	0.075	0.069	0.4724	0.4459	n.s.	0.0288	0.032 (0.420)	-0.03388	-1.491	NO	0.010	no
L1d/S3d	diag	69	3.338	2.851	0.0022	0.0054	suggestive	1.0432	1.402 (0.437)	0.07494	2.662	yes	0.870	no
L28d/S2d	diag	69	0.872	0.937	0.1744	0.1758	n.s.	0.2497	0.366 (0.421)	0.00329	0.182	yes	0.060	no
L28d/S3d	diag	69	1.416	1.393	0.0818	0.0804	n.s.	0.3889	0.595 (0.423)	0.01942	1.072	yes	0.142	no
L3d/S2d	diag	69	2.766	2.378	0.0087	0.0192	suggestive	1.0109	1.161 (0.505)	0.04401	2.116	NO	0.653	no
L3d/S3d	diag	69	4.663	5.015	0.0000	0.0002	sig	1.0807	1.958 (0.452)	0.08106	4.743	yes	0.985	no
L56d/S2d	diag	69	1.217	0.990	0.1610	0.1466	n.s.	0.3780	0.511 (0.522)	0.00931	0.425	NO	0.099	no
L56d/S3d	diag	69	1.075	0.891	0.1866	0.1658	n.s.	0.3159	0.451 (0.513)	0.01094	0.528	NO	0.081	no
L5d/S2d	diag	69	2.508	2.588	0.0048	0.0108	suggestive	0.9616	1.053 (0.429)	0.04526	2.060	yes	0.558	no
L5d/S3d	diag	69	2.857	2.660	0.0039	0.0030	suggestive	0.8167	1.200 (0.432)	0.05060	2.187	yes	0.675	no
L7d/S2d	diag	69	2.097	2.327	0.0100	0.0116	suggestive	0.6428	0.881 (0.427)	0.03529	1.691	yes	0.372	no
L7d/S3d	diag	69	1.925	2.102	0.0178	0.0110	suggestive	0.5245	0.808 (0.426)	0.03546	2.082	yes	0.330	no

7.4 Full Stage-4 tables

Comparator L28d/S1d (academic 4-week) — gross vs net, IS vs OOS:

segment	gross Sharpe	net Sharpe	net ann ret	net ann vol	Sortino	max DD	Calmar	hit rate	ann turnover
in-sample	0.073	-0.031	-0.0134	0.4314	-0.030	-0.6809	-0.020	0.559	23.90
out-of-sample	-0.158	-0.270	-0.0818	0.3031	-0.283	-0.5168	-0.158	0.433	26.57

L28d works only gross of costs: net Sharpe is negative in both segments.

Additional net metrics (total return, avg win / loss):

spec	segment	total return	avg win	avg loss
Primary L5d_S1d	in-sample	1.7225	0.08584	-0.06745
Primary L5d_S1d	out-of-sample	0.6266	0.07509	-0.05136
Comparator L28d_S1d	in-sample	-0.4518	0.06534	-0.08526
Comparator L28d_S1d	out-of-sample	-0.2687	0.06283	-0.05991

2×-costs sensitivity (in-sample, primary L5d/S1d):

segment	gross Sharpe	net Sharpe	net ann ret	net ann vol	Sortino	max DD	Calmar	hit rate	ann turnover
1x costs	0.789	0.664	0.2490	0.3752	0.756	-0.4851	0.513	0.574	24.31
2x costs	0.789	0.552	0.2072	0.3753	0.610	-0.5219	0.397	0.574	24.32

Doubling fees + slippage cuts net Sharpe 0.664 → 0.552 (same spec; not a re-selection).